

Zeeshan Samana

Senior AI/ML & Generative AI Engineer

LLMs | Agentic AI | RAG | Python | Backend Systems

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PROFESSIONAL SUMMARY

Senior AI/ML and Python Engineer with 9+ years of experience designing and building intelligent software systems, Generative AI applications, Large Language Model (LLM) solutions, agentic AI workflows, and scalable backend platforms across Industrial IoT, Smart Buildings, Legal Technology, Life Sciences, and Enterprise Software domains. Expertise in Python, FastAPI, Django, Machine Learning, Deep Learning, NLP, Retrieval-Augmented Generation (RAG), AI Agents, LangGraph, LangChain, AWS Bedrock, TensorFlow, PyTorch, Hugging Face, OpenSearch, Pinecone, Neo4j, and vector databases. Proven experience developing production-grade AI applications that combine structured and unstructured data, knowledge retrieval, reasoning workflows, and intelligent automation to solve complex business problems. Experienced in building LLM applications, AI agents, multi-agent systems, semantic search platforms, vector-based knowledge retrieval solutions, intelligent automation systems, and enterprise AI applications using modern Generative AI technologies. Strong background in backend engineering, distributed systems, microservices architecture, API development, event-driven applications, and cloud-native software development on AWS, Azure, and Google Cloud Platform. Experienced in building end to end AI products from data processing and model development through deployment, monitoring, evaluation, and continuous improvement.

TECHNICAL SKILLS

- **Artificial Intelligence & Machine Learning:** Generative AI, Large Language Models (LLMs), Agentic AI, Retrieval-Augmented Generation (RAG), Machine Learning, Deep Learning, NLP, Computer Vision, Knowledge Graphs, Prompt Engineering, Fine-Tuning, Model Evaluation, AI Agents, Multi-Agent Systems
- **AI/ML Frameworks & Libraries:** LangGraph, LangChain, PydanticAI, AWS Bedrock, Hugging Face Transformers, TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLO, SpaCy, MLflow
- **Programming Languages:** Python, SQL, JavaScript, TypeScript, Bash
- **Backend Development:** FastAPI, Django, Flask, REST APIs, GraphQL, gRPC, Microservices Architecture, Event-Driven Systems, Distributed Systems
- **Data Engineering & Processing:** Apache Spark, Kafka, Airflow, Dask, Pandas, NumPy, ETL/ELT Pipelines, Data Modeling, Data Analytics
- **Databases & Search Technologies:** PostgreSQL, MySQL, MongoDB, Redis, Snowflake, OpenSearch, Elasticsearch, Pinecone, FAISS, Neo4j, AWS Neptune, Vector Databases
- **Cloud & Infrastructure:** AWS, Microsoft Azure, Google Cloud Platform (GCP), Docker, Kubernetes, Amazon EKS, Azure AKS, Google GKE, Terraform
- **DevOps & MLOps:** CI/CD, GitHub Actions, Jenkins, GitLab CI/CD, Kubeflow, MLflow, Argo Workflows, Model Deployment, Model Monitoring
- **Observability & Monitoring:** Prometheus, Grafana, Datadog, CloudWatch, OpenTelemetry, Distributed Tracing
- **Software Engineering:** System Design, Software Architecture, Design Patterns, API Design, Scalability, Performance Optimization, Security Best Practices, Agile Development

WORK EXPERIENCE

Senior AI/ML Platform & Generative AI Engineer

Johnson Controls | *Milwaukee, WI (Remote)*

Jul 2022 – Present

- Architect and lead development of enterprise AI applications using Python, FastAPI, Django, React.js, AWS, and Azure to support intelligent building operations, energy optimization, predictive maintenance, and connected IoT ecosystems.
- Delivered production-grade Generative AI solutions leveraging LangGraph, PydanticAI, AWS Bedrock, and agent orchestration frameworks to automate complex operational workflows and accelerate field-service decision making.
- Established retrieval-augmented generation (RAG) platforms integrating OpenSearch, Pinecone, Azure AI Search, and FAISS, enabling contextual knowledge access across engineering documentation, maintenance records, operational data, and enterprise knowledge repositories.
- Designed knowledge graph architectures using Neo4j and AWS Neptune to model equipment relationships, infrastructure dependencies, and failure propagation patterns for advanced root-cause analysis and intelligent reasoning.
- Developed machine learning solutions using TensorFlow, PyTorch, and Scikit-learn to support predictive maintenance, anomaly detection, asset health monitoring, operational forecasting, and condition-based maintenance initiatives.
- Built computer vision applications using OpenCV, YOLO, and multimodal AI models to automate equipment inspection, workplace safety monitoring, defect identification, and visual intelligence workflows.
- Engineered large-scale data processing frameworks using Apache Spark, Dask, Snowflake, and Kafka to transform telemetry streams into actionable insights for AI-driven automation and advanced analytics.
- Designed scalable backend services, REST APIs, gRPC services, and event-driven architectures that enabled seamless integration between AI systems, enterprise applications, IoT platforms, and operational workflows.

- Developed enterprise AI agents and multi-agent workflows using LangGraph, AWS Bedrock Agents, and tool-calling frameworks that automated complex business processes, orchestrated actions across systems, and enhanced operational efficiency.
- Built advanced Retrieval-Augmented Generation (RAG) systems combining vector databases, semantic search, knowledge graphs, and large language models to deliver accurate, context-aware responses across technical documentation, operational records, and enterprise knowledge repositories.
- Designed and evaluated LLM-powered applications using prompt engineering, model evaluation frameworks, guardrails, and AI observability practices to improve response quality, reduce hallucinations, and ensure reliable production AI experiences.
- Spearheaded the development of AI-powered copilots and conversational assistants that integrated enterprise knowledge, operational data, and business workflows to enhance decision-making, accelerate issue resolution, and improve user productivity across engineering and operations teams.

Senior AI Systems Engineer

Thomson Reuters | *Eagan, MN (Remote)*

Nov 2020 – Jul 2022

- Designed and developed AI-powered backend platforms using Python, FastAPI, Flask, and cloud-native services supporting legal research, regulatory intelligence, compliance analytics, and risk-management solutions.
- Built semantic retrieval and knowledge discovery systems utilizing OpenSearch, Pinecone, FAISS, and Azure AI Search to improve access to legal, regulatory, and compliance information.
- Delivered advanced natural language processing solutions using Hugging Face Transformers and SpaCy for document understanding, entity extraction, classification, enrichment, and knowledge organization.
- Developed intelligent workflow automation services that combined machine learning models, search technologies, and business-rule engines to streamline legal and compliance processes.
- Implemented scalable data engineering pipelines using Apache Spark, Kafka, and distributed processing frameworks to support AI training, document analytics, and enterprise search workloads.
- Enhanced ranking, classification, and anomaly-detection capabilities through machine learning model development, feature engineering, experimentation, and continuous optimization practices.
- Designed API-driven microservices architectures that enabled integration between AI services, enterprise platforms, internal data sources, and customer-facing applications.
- Established cloud-native deployment standards using Docker, Kubernetes, GKE, and automated delivery pipelines to improve platform scalability, reliability, and operational consistency.
- Leveraged GCP services including BigQuery, Cloud Storage, Pub/Sub, and Kubernetes to support both real-time and batch AI processing workloads.
- Collaborated closely with legal domain experts, product teams, and engineering leadership to translate complex business requirements into scalable AI-enabled products.
- Introduced model governance, experimentation workflows, and reproducible ML development practices using MLflow and Airflow to strengthen platform maturity.
- Built intelligent knowledge-discovery solutions that combined natural language processing, semantic search, and machine learning techniques to help legal and compliance professionals efficiently uncover insights from large volumes of structured and unstructured information.

Senior Software Engineer (Python)

Veeva Systems | *Pleasanton, CA*

Oct 2018 – Nov 2020

- Designed and developed backend platforms using Python, Django, and service-oriented architectures supporting regulatory, clinical, quality, and content-management solutions within highly regulated environments.
- Built intelligent document-processing pipelines combining OCR, NLP, and machine learning technologies to automate extraction, classification, validation, and enrichment of regulatory content.
- Developed machine learning solutions using Scikit-learn and TensorFlow to improve data quality, anomaly detection, document categorization, and validation workflows.
- Implemented enterprise search and knowledge retrieval capabilities using Elasticsearch and semantic search techniques to enhance content discovery and regulatory information access.
- Created scalable REST APIs and microservices enabling seamless integration between Veeva Vault, CRM systems, regulatory applications, and external business platforms.
- Developed model-serving frameworks and inference services using FastAPI to support operational machine learning workloads within business-critical applications.
- Automated data processing and machine learning workflows using Airflow and MLflow, improving reproducibility, governance, and deployment consistency.
- Collaborated with compliance, quality assurance, product management, and engineering teams to ensure solutions aligned with industry regulations and operational requirements.
- Strengthened platform reliability through performance optimization, testing automation, code quality initiatives, and service modernization efforts.
- Played a key role in advancing intelligent automation capabilities across regulated business processes while maintaining compliance and auditability standards.
- Developed data validation and workflow automation services that streamlined regulatory submission processes, improved data integrity, and enabled efficient collaboration across clinical, quality, and compliance teams.
- Designed scalable event-driven integrations using Python, message queues, and REST APIs to synchronize data between Veeva Vault, enterprise applications, and external regulatory systems, improving operational efficiency and system interoperability.

- Developed and maintained backend applications using Python, Django, Flask, and MySQL supporting digital commerce, product management, and customer engagement initiatives.
- Designed scalable RESTful APIs and backend services enabling seamless integration between web applications, business systems, and third-party platforms.
- Implemented Elasticsearch-powered search experiences that enhanced product discovery, catalog navigation, and customer self-service capabilities.
- Built secure payment integrations with major payment gateways, improving transaction processing and supporting online purchasing workflows.
- Developed data-processing solutions using Pandas and NumPy to support reporting, business intelligence, operational analytics, and customer behavior analysis.
- Created recommendation and personalization features using machine learning techniques to improve content relevance and user experience.
- Applied natural language processing techniques for content classification, search optimization, and customer interaction analysis.
- Implemented authentication and authorization frameworks using OAuth and JWT to strengthen application security and access control.
- Introduced CI/CD automation, testing pipelines, and deployment workflows using Jenkins and GitLab CI, improving release consistency and development efficiency.
- Partnered with product owners, designers, and business stakeholders to deliver scalable backend solutions that supported digital transformation and platform growth.
- Collaborated with product managers, UX teams, and business stakeholders to deliver customer-facing features that enhanced user experience, streamlined purchasing workflows, and supported the company's digital commerce initiatives.
- Improved application reliability and maintainability by implementing automated testing, code quality standards, and performance optimization practices, enabling faster feature delivery and long-term platform scalability.

PROJECTS

AI-Driven Predictive Maintenance for Industrial IoT

- Designed an AI-powered predictive maintenance platform that analyzed industrial IoT sensor data using machine learning and deep learning models for anomaly detection, equipment health monitoring, and failure prediction. Built scalable real-time data pipelines with Kafka, Spark, and Airflow, and deployed cloud-native inference services on Google Cloud Vertex AI to support proactive maintenance operations.

Distributed RAG & Semantic Search Platform

- Developed a large-scale Retrieval-Augmented Generation (RAG) platform combining vector search, semantic retrieval, and intelligent re-ranking to enable contextual search and question-answering across enterprise knowledge repositories. Built scalable ingestion and embedding pipelines with Kafka and Spark while serving low-latency AI search APIs through FastAPI and Redis

Enterprise MLOps & Feature Engineering Platform

- Built an enterprise MLOps platform that standardized model development, experiment tracking, deployment automation, feature management, and model monitoring across multiple machine learning teams. Implemented automated CI/CD workflows, drift detection, and observability capabilities to accelerate the delivery of production-ready AI solutions.

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate
- AWS Certified Solutions Architect – Professional (SAP-C02)
- Microsoft Certified: Azure AI Engineer Associate (AI-102)

EDUCATION

Bachelor of Computer Science

Allama Iqbal Open University